SPE-229177-MS

Feasibility of Converting Condensate Gas Reservoirs into Compressed Air Energy Storage Facilities in the Later Stage of Development

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

Recently, the rapid development of renewable energy has highlighted the crucial role of energy storage in addressing temporal mismatches between energy supply and demand. Compressed Air Energy Storage (CAES), as a long-duration energy storage method, offers significant advantages. This paper proposes to conduct feasibility assessments for implementing air energy storage in larger-scale condensate gas reservoirs, aiming to meet future demands for more extended and stable energy storage solutions.

Based on the requirements for CAES facilities, this study takes a condensate gas reservoir as an example to evaluate reservoir characteristics, gas storage capacity, and caprock stability. A subsurface model was established to simulate the processes of gas injection for energy storage and gas production for power generation. The distribution patterns and saturation variations of residual in-situ hydrocarbons, injected air, and formation water were analyzed. Based on a conservative gas storage capacity assessment, an economic evaluation of retrofitting the CAES facility was preliminarily conducted, with predictions of economic benefits and development prospects.

First, studies reveal that existing oil-gas reservoirs generally exhibit superior geological conditions with substantial gas storage capacities, where effective storage spaces typically exceed 100 million cubic meters. The caprock sealing capacity formed through long-term geological evolution demonstrates excellent containment properties. Second, numerical simulations indicate an energy storage efficiency of 80% under aquifer-free conditions. Effective storage capacity decreases with increasing aquifer presence and enhanced formation heterogeneity, suggesting preferential implementation in condensate gas reservoirs with weak aquifer systems. Third, Cyclic gas injection/production enables simultaneous energy storage and theoretically enhanced oil/gas recovery by approximately 8 percentage points, generating dual revenue streams. Fourth, A lifecycle economic model for a 20-million-m^3 storage system yields an internal rate of return (IRR) of 14.3% with a payback period of 6.8 years, confirming promising long-term economic viability.

In gas and condensate gas reservoir enrichment zones, when integrated with ultra-high voltage (UHV) transmission networks, these formations could establish regional clean energy dispatch centers, enabling

long-term sustainable utilization of mineral resources. This configuration demonstrates significant potential for achieving green energy transition and creating substantial prospects for future energy infrastructure development.

Introduction

Global Renewable Energy Landscape and the Energy Storage Imperative

The global renewable energy sector, particularly solar photovoltaic and wind power, is experiencing rapid growth. As detailed in the International Energy Agency's *(IEA)* "Renewables 2024" report, projections indicate a substantial increase in global renewable energy installed capacity by 2030, with this expansion equivalent to the total installed power capacity of current major global economies. China is poised to contribute nearly 60% of this growth. From the perspective of power generation technology, it is anticipated that by 2030, solar photovoltaic power generation alone will account for 80% of the global increase in renewable energy capacity, and renewable energy sources will collectively produce nearly half of the world's electricity, with wind and solar photovoltaic generation's share doubling to 30%. Despite this rapid expansion, renewable energy generation faces inherent challenges such as intermittency, periodicity, and instability *(Hannan et al., 2023)*. This has led to a notable increase in curtailment rates, the proportion of unused renewable energy generation, which has reached approximately 10% in several countries. Addressing this issue necessitates enhanced grid flexibility and robust energy integration measures. The intrinsic variability and unpredictable nature of renewable energy sources, while offering immense environmental benefits, concurrently impose significant operational challenges on existing power grids. This inherent characteristic directly drives a surge in demand for grid peak-shaving capabilities and, consequently, an urgent requirement for large-scale, long-duration energy storage technologies. These dynamic positions energy storage not merely as an auxiliary component, but as a foundational prerequisite for the successful integration and widespread expansion of renewable energy within the global power system *(IEA, 2024)*.

Renewable energy's intermittent, cyclical, and unstable nature has intensified grid peak-shaving demands. Developing new power systems reliant on renewables is a key strategy for increasing their share in end-use electricity supply. The energy transition has created an urgent need for large-scale, long-duration energy storage technologies *(Mohammad Amira et al., 2023; Zhong et al., 2024)*. Energy storage, a cornerstone of modern power systems *(Yang et al., 2022)*, enables time-shifted energy storage and release, smoothing renewable output, balancing peak and off-peak grid loads, managing user-side energy, and supporting the broader adoption of renewables.

Role of Long-Duration Energy Storage in Modern Power Systems

Energy storage technologies are pivotal for achieving a stable and efficient new power system. They facilitate smooth output from renewable sources, enable peak shaving and valley filling in the power grid, support user-side energy management, and contribute to balancing overall power load. The fundamental principle of energy storage lies in enabling the temporal and spatial relocation of energy *(LI Jianlin 2023; SUN Yushu 2020)*, thereby enhancing energy controllability.

Long-duration energy storage (LDE), typically defined as technologies capable of storing energy for more than 4 hours, is crucial for cross-day, cross-month, and even cross-seasonal charge and discharge cycles, ensuring the long-term stability of the power system. Current LDE technologies encompass three main categories: mechanical energy storage (e.g., pumped hydro storage, CAES), thermal energy storage (e.g., molten salt thermal storage), and chemical energy storage (e.g., lithium-ion, sodium-ion, and flow batteries). Among these, pumped hydro storage (PHS) and CAES are distinguished by their large storage scale, extended-release times, and long operational lifespans.

Overview of CAES Technology

CAES technology originated from the development of gas turbine technology *(CHEN Haisheng, 2013)*, with its earliest conceptualization attributed to STAL LAVAL in 1949 *(Ter-Gazarian A, 1994)*. Key advantages of CAES include lower construction costs, environmental friendliness, and a long operational lifespan. The system operates by converting electrical energy into potential and thermal energy during storage and subsequently releasing electrical energy through a reverse conversion process, making it an optimal choice for future large-scale, long-duration energy storage applications.

While pumped hydro storage (PHS) is a mature and large-scale LDE technology, its stringent geographical requirements, demanding the construction of two reservoirs at different altitudes *(RAO Hu, 2024; CAI Wenfang, 2025)*, impose significant limitations on its widespread deployment. This strict site selection criterion represents a primary constraint for the broader adoption of PHS. In contrast, CAES, particularly when leveraging underground geological formations, offers considerably greater site selection flexibility. This adaptability positions CAES as a more versatile and strategically superior option for future large-scale, long-duration energy storage infrastructure, overcoming the inherent geographical constraints that limit PHS.

Global and Domestic CAES Development Status

Globally, significant milestones in CAES development include the Huntorf CAES Plant in Germany, which became the world's first commercially operational CAES plant in 1978 *(Heidar Jafarizadeh, 2020; Friederike Kaiser, 2018; Zhang Xinjing, 2024)*. This plant features two underground gas storage caverns with a total volume of 310,000 cubic meters, operating within a pressure range of 43-70 bar, and providing an electrical output power of 290 MW with a capacity of 870 MWH. However, its reliance on natural gas combustion results in a carbon emission intensity of 412 gCO_2/kWh.

The second globally operational CAES plant is the McIntosh CAES Plant in Alabama, USA, commissioned in 1991 *(GUO Zuogang, 2019)*. This plant improved upon the Huntorf design by incorporating an expander exhaust waste heat recovery system, which transfers heat from the expander exhaust to the compressed air, thereby reducing natural gas consumption. The McIntosh plant has a compressor unit power of 50 MW, an expander-generator output of 110 MW, and a total underground cavern volume of 5.6×10^5 m^3 at a depth of 450 m, capable of continuous energy storage for 41 hours and continuous power output for 26 hours.

China's CAES technology is advancing rapidly, primarily focusing on non-combustion CAES, establishing a multi-type, large-scale application pattern with trends towards high efficiency and standardization *(China New Energy Storage Development Report 2025)*. By the end of 2024, CAES constituted 1.0% (approximately 737.6 MW) of China's total new energy storage installed capacity, ranking as the second-largest technology route after lithium-ion batteries (96.4%). Notably, 2024 saw an addition of 11.78 GW in CAES installed capacity, marking a 70-fold year-on-year increase. With 13 projects under construction (3.96 GW) and 13 planned projects (5.5 GW), the commercialization process is accelerating. Key representative projects include the Shandong Feicheng 300MW/1800MWh plant, which is the world's first 300 MW-class commercial power station with a system efficiency of 72.1% and an annual power generation of 600 million kWh. The Shandong Tai'an 350MW project stands as the world's largest single-unit salt cavern energy storage power station, designed for over 70% efficiency and an annual power generation of 460 million kWh, achieving zero carbon emissions by utilizing abandoned salt cavern resources. Additionally, the Hubei Yingcheng 300MW project utilizes a "molten salt + pressurized hot water" thermal storage solution, achieving an efficiency of 70%.

Presently, most air compression facilities primarily utilize underground salt caverns *(Guo Chaobin, 2021; GUO Dingzhang, 2021)*. The abundance of salt cavern resources can significantly reduce construction costs, making them the mainstream for current commercial projects. Exploration into artificial cavern gas storage, involving the modification of mines or the construction of new underground caverns, is ongoing to mitigate geographical limitations. However, there are no operational projects in China yet, with this technology remaining in the experimental stage *(DU Dongmei, 2024; XU Yingjun, 2022)*.

Rationale for Utilizing Condensate Gas Reservoirs for CAES

The trajectory of CAES is towards large-scale development, aiming to reduce unit investment costs and facilitate widespread adoption, aligning with the requirements of evolving new power systems. Existing conventional gas storage facilities, including above-ground installations, natural salt caverns, and artificial caverns, are increasingly insufficient to meet the escalating demand for larger storage volumes required by next-generation, larger-scale CAES power plants.

This paper uniquely addresses this challenge by proposing and conducting a feasibility assessment for implementing air energy storage in larger-scale condensate gas reservoirs. This innovative approach seeks to fulfill future demands for more extended and stable energy storage solutions, leveraging existing geological infrastructure. The strategic shift from constructing new, purpose-built CAES caverns (like salt or artificial caverns) to repurposing existing, depleted condensate gas reservoirs signifies a profound paradigm change in energy infrastructure development. This approach not only resolves the critical issue of large-scale storage capacity but also offers an invaluable "second life" for mineral resource assets, promoting principles of circular economy and sustainable resource management. By leveraging inherent geological advantages and existing subsurface infrastructure, this pathway presents a more resource-efficient and environmentally beneficial route for advancing large-scale energy storage.

Methodology

This study's primary focus is a subsurface analysis, evaluating the potential of condensate gas reservoirs as CAES facilities. This includes assessing their scale, predicting changes in saturation, analyzing fluid composition variations, and estimating overall energy storage capacity.

Geological Characterization and Evaluation

A comprehensive approach combining geological surveys and engineering experiments was used to assess the storage potential and geological risks of condensate gas reservoirs. Core property tests, well log interpretations, and 3D seismic data inversion quantitatively characterized structural morphology, reservoir porosity, and permeability distribution. Caprock evaluation involved breakthrough pressure tests and triaxial rock mechanics tests to assess plastic deformation capacity. Caprock and interlayer sealing integrity, critical for storage feasibility, were analyzed based on thickness, morphology, composition, and permeability. Geomechanical simulations predicted caprock sealing integrity under cyclic injection-production loads, identifying fault reactivation risks. A storage potential grading map and caprock failure probability model were developed to guide site selection. Volumetric methods calculated pore volumes and geological gas reserves at the end of hydrocarbon production, adjusting effective working gas volume based on injection-production pressure windows (minimum/maximum operating pressures), validated by numerical simulation.

Reservoir Simulation of Gas Storage Process

Based on a detailed geological model, a multi-component thermodynamic flow coupled numerical model (Fig.1) is constructed to accurately represent the complex subsurface processes. The model is used to simulate the complete working cycle of CAES, providing insights into the dynamic behavior of the reservoir.

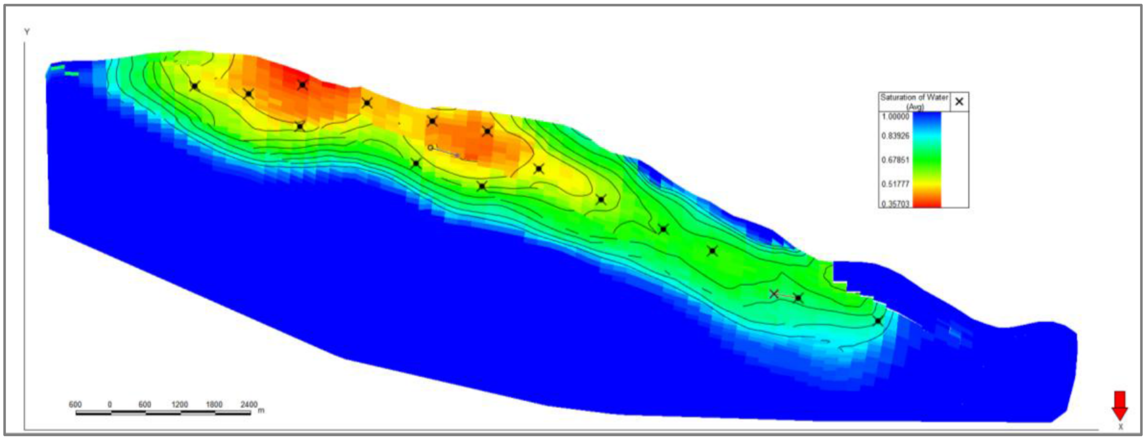

Figure 1—Original Gas-Water Distribution in Y Gas Reservoir

The simulation tracks the spatio-temporal evolution of component saturation fields, including residual condensate oil, injected air, and formation water. This analysis aims to reveal the anti-evaporation behavior of residual condensate oil under pressure oscillations and the dynamics of formation water invasion. Furthermore, it allows for the estimation of natural gas and condensate oil recovery rates, calculation of gas storage capacity and working gas volume, and evaluation of changes in hydrocarbon gas molar content. The insights derived from these simulations provide crucial theoretical support for optimizing well patterns and implementing effective safety control measures.

Energy Storage Capacity and Energy Density Calculation

Air serves as the fundamental energy storage medium, making its thermodynamic properties and behavior critical for the analysis and calculation of CAES systems. The paper acknowledges various thermodynamic processes relevant to CAES, including adiabatic, isochoric, isobaric, and isothermal processes. For calculating the theoretical maximum energy, the study adopts the assumption of ideal isothermal expansion, which is a commonly used simplified model for large underground gas storage facilities due to their large thermal mass. For a fixed volume (V) gas storage cavern, the stored or released energy (E) is given by the following formula:

$$E = V \cdot P_{\min} \cdot \ln\left(\frac{P_{\max}}{P_{\min}}\right) \tag{1}$$

Where: P_{min} is the absolute outlet pressure at the end of expansion (Pa); P_{max} is the initial absolute pressure of gas storage (Pa); V is the gas storage volume (m^3); and *ln* is the natural logarithm.

Power generation (P) is defined as the energy released divided by the power generation time (t), in hours:

$$P = \frac{E}{t} \tag{2}$$

Energy density, the energy stored per unit volume (J/m^3), is calculated similarly:

$$\eta = \frac{E}{V} = P_{min} \cdot \ln\left(\frac{P_{max}}{P_{min}}\right) \tag{3}$$

Economic Benefit Estimation

Based on the potential gas storage capacity, a simplified cost-benefit prediction model is established. Initial investment costs encompass a range of expenditures, including new drilling operations, workovers and modifications to existing wells, upgrades to gas injection facilities, and the installation of surface power generation facilities. Recurring operating costs include electricity consumption for compression (accounted for with time-of-use pricing), equipment maintenance, water treatment, and labor expenses.

Primary revenue streams are derived from the arbitrage of peak-valley electricity price differences, calculated as daily cycle times multiplied by single discharge volume and the price difference. Potential carbon trading benefits are also considered as a supplementary revenue source. The core dynamic economic indicators used for evaluation are Net Present Value (NPV) and Internal Rate of Return (IRR). These are complemented by an analysis of the payback period and the unit energy storage cost (CNY/kWh) to provide a comprehensive financial assessment.

Results and Discussion

Geological Adaptability and Storage Potential (Case Study: Y Condensate Gas Reservoir)

The Y gas reservoir exhibits an original reservoir pressure of 56 MPa. The caprock seal integrity evaluation indicates that the caprock's fracture pressure is reached when the fluid pressure exceeds the original formation pressure by 15 MPa (i.e., at 71 MPa). The central area of the gas reservoir is identified as the primary risk zone for potential gas leakage. Fault seal integrity assessment suggests that faults may be activated when the fluid pressure surpasses the formation pressure by 7.5 MPa (i.e., at 63.5 MPa). Consequently, the upper operating pressure limit for the gas storage facility should not exceed 63.5 MPa. This limit is significantly higher than the typical upper pressure limit of 7 MPa for existing salt cavern type gas storage facilities, implying an extremely low probability of gas leakage during operation. It is recommended that the actual operating pressure remains below the original formation pressure to ensure maximum safety. The substantial difference in operating pressure limits (63.5 MPa for gas reservoirs versus 7 MPa for salt caverns) is a critical advantage. This not only provides a significantly larger safety margin against gas leakage but also fundamentally enhances the volumetric storage efficiency. The inherent geological integrity and deep burial of natural reservoirs allow for much higher operating pressures, directly translating to a greater quantity of energy stored per unit volume. This capability is pivotal for achieving the large-scale, high-density energy storage required for grid-level applications, surpassing the inherent limitations of shallower, lower-pressure salt caverns.

The geological natural gas reserves of the Y gas reservoir are approximately 29 billion cubic meters, which translates to an underground volume of about 100 million cubic meters. Even at the end of its development, when the abandonment pressure of 22 MPa is reached (Fig.2), the underground gas storage space retains a substantial volume of 20 million cubic meters (Fig.3). This volume is tens of times larger than the typical scale of current salt cavern type gas storage facilities. At this abandonment stage, precipitated condensate oil remains in the pores, with a saturation ranging from 4-9% (Fig.4), accounting for approximately 4.2 million tons of residual condensate oil reserves in the reservoir.

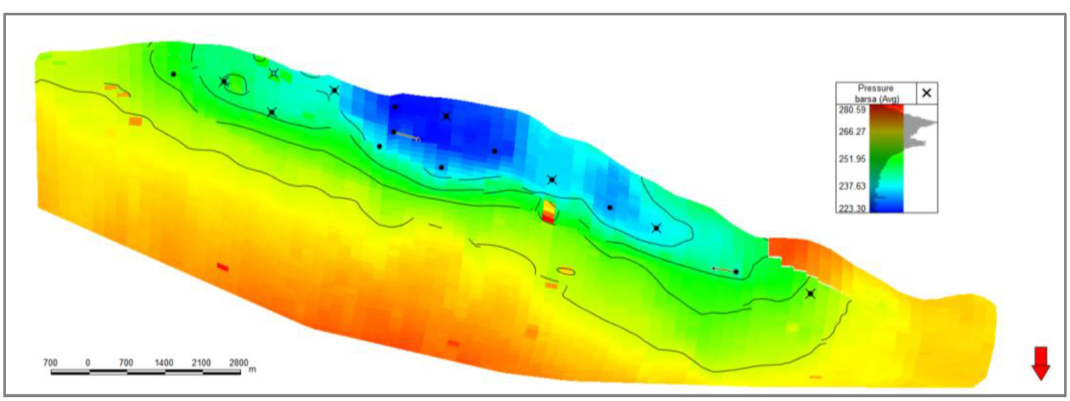

Figure 2—Pressure Distribution in Y Gas Reservoir at Abandonment

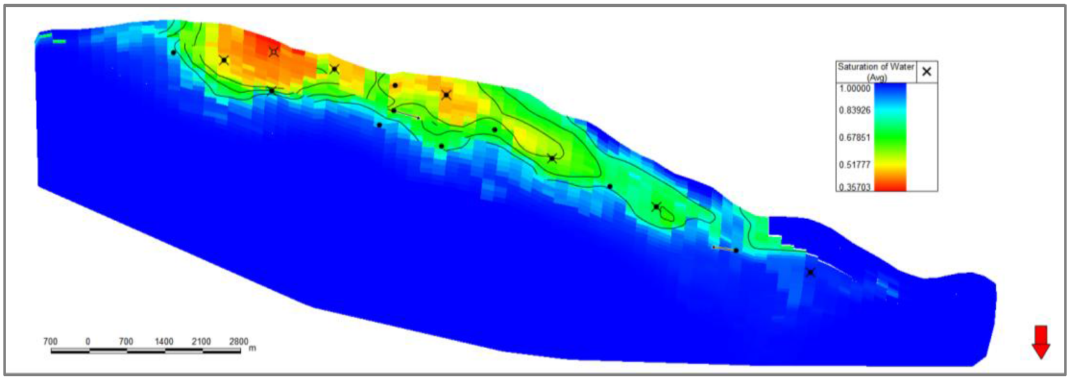

Figure 3—Gas-Water Distribution in Y Gas Reservoir at Abandonment

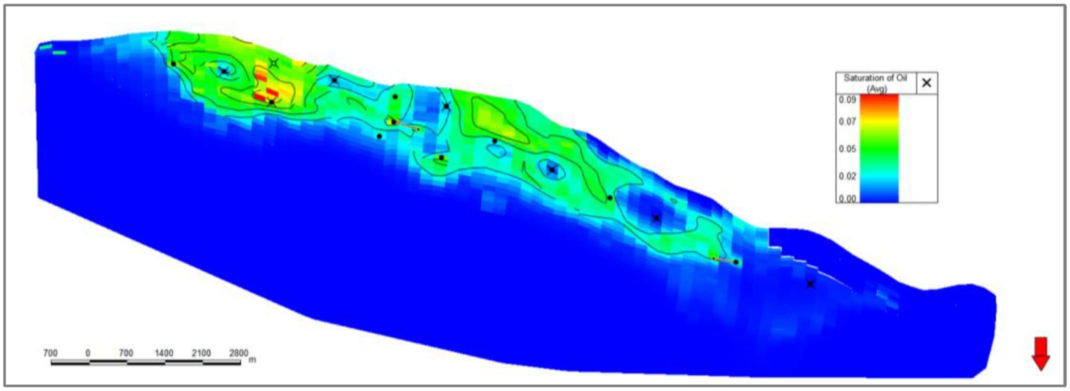

Figure 4—Condensate Oil Saturation Distribution in Y Gas Reservoir at Abandonment

Impact of Geological Conditions on CAES Performance

Numerical simulations reveal a significant impact of geological conditions on CAES performance. Energy storage efficiency reaches 80% in the absence of natural aquifers. However, effective gas storage space decreases with increasing aquifer presence and enhanced formation heterogeneity. The primary influencing factors are aquifer energy and heterogeneity. Larger aquifers tend to cause uneven water advancement during high-rate energy release, which can severely lead to water breakthrough in gas production wells. Stronger heterogeneity and greater permeability contrast accelerate the fingering of water along high-permeability layers, further impeding efficient gas storage and recovery.

Based on these findings, a reservoir screening priority is proposed: weak aquifer systems (with a volume multiple of less than 3) are preferred over constant volume systems, which are in turn preferred over strong aquifer systems. It is strongly recommended to avoid fractured reservoirs and to prioritize porous, weak aquifer condensate gas reservoirs for CAES implementation. The detailed analysis of how aquifer presence and heterogeneity impact CAES performance provides critical operational considerations beyond mere geological assessment. It establishes that the specific type of condensate gas reservoir (e.g., weak versus strong aquifer, porous versus fractured) is a crucial determinant of long-term CAES efficiency and operational stability. This directly influences optimal site selection, well placement strategies, and the design of measures to mitigate risks such as water breakthrough, thereby moving the analysis from basic feasibility to practical engineering and operational optimization.

Energy Storage Capacity and Power Generation Analysis

For calculation purposes, the upper limit of gas storage operating pressure is assumed to be the original formation pressure of 56 MPa, and the lower limit is the abandonment pressure of 22 MPa. A conservative working gas volume of 20 million cubic meters at abandonment pressure is used. Based on the isothermal process energy calculation method, the estimated released energy is approximately 4×10^{14} J, which converts to about 114 GWh. This capacity is remarkably 60 times the power generation scale of currently operating CAES power plants, indicating a scale equivalent to a very large energy storage power plant. If this substantial energy is released over a 24-hour period, the resulting power output would be 4.76 GW. This is 13 times the power of the world's largest salt cavern CAES power plant, demonstrating its suitability for grid-scale applications.

Due to the significantly greater burial depth of gas reservoirs compared to salt caverns, their operating pressure is substantially higher. This characteristic profoundly enhances the energy storage capacity per unit volume, leading to a much greater stored energy density. For comparison, the energy density of the Huntorf plant in Germany is approximately 0.8 kWh/m^3. In contrast, the energy density for gas reservoir type storage, as analyzed in this study, is 5.7 kWh/m^3, which is 7 times that of typical salt cavern types. The seven-fold increase in energy density for gas reservoir-based CAES compared to salt caverns is not merely a technical metric; it carries profound economic and strategic implications.

Underground gas storage serves as energy storage facilities, thereby achieving the goal of converting large-scale, unstable renewable energy into stable, controllable, and commercial electricity (Fig. 5).

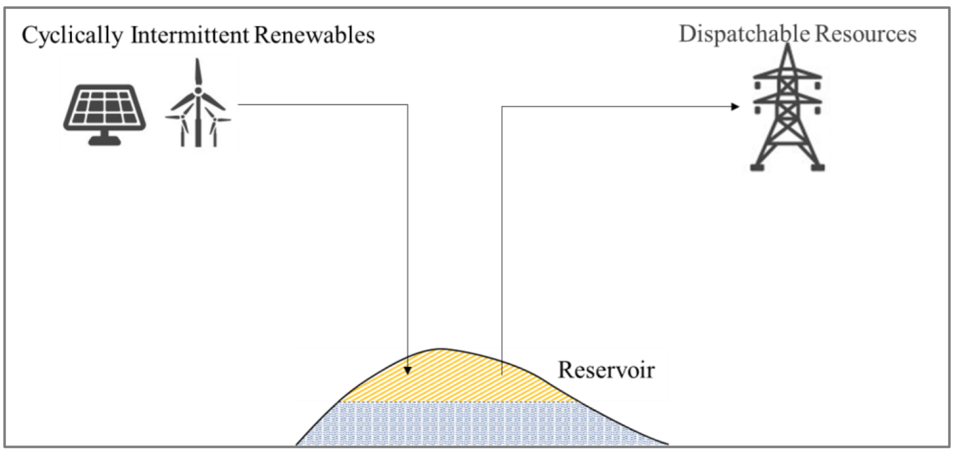

Figure 5—Schematic Diagram of Underground Gas Reservoir CAES Effect

Potential for Enhanced Hydrocarbon Recovery

A critical safety consideration when converting to an air energy storage facility is the presence of oxygen in the injected air. Therefore, we must conduct comprehensive preparatory work before transitioning to air energy storage mode to mitigate risks of oxygen-hydrocarbon mixtures. During this preparatory phase, specific designs can be implemented to enhance the recovery of residual natural gas and condensate oil, transforming a safety challenge into an economic opportunity.

The envisioned process for hydrocarbon displacement and safety assurance includes three steps:

1. Nitrogen Injection: Initially, nitrogen injection is performed at top well locations. This achieves multiple objectives: restoring formation pressure, pushing back invading formation water, and expanding the effective gas storage capacity. Critically, as pressure increases, a portion of the already precipitated condensate oil will undergo vaporization, returning to the gas phase, with the saturation of condensate decreasing from 9% to 4% (Fig. 6).
2. Peripheral Production: Subsequently, the gas-oil ratio and condensate oil content in peripheral production wells are monitored. If production is deemed economically viable, these wells can be opened for production. This continues until nitrogen breakthrough occurs and the hydrocarbon gas content in the produced stream significantly decreases.

3. Air Storage Mode Initiation: After several cycles of nitrogen injection and hydrocarbon production, the total hydrocarbon component content throughout the reservoir is reduced to a predefined safe limit (e.g., methane molar content less than 5%). Only then does the air energy storage mode operation commence.

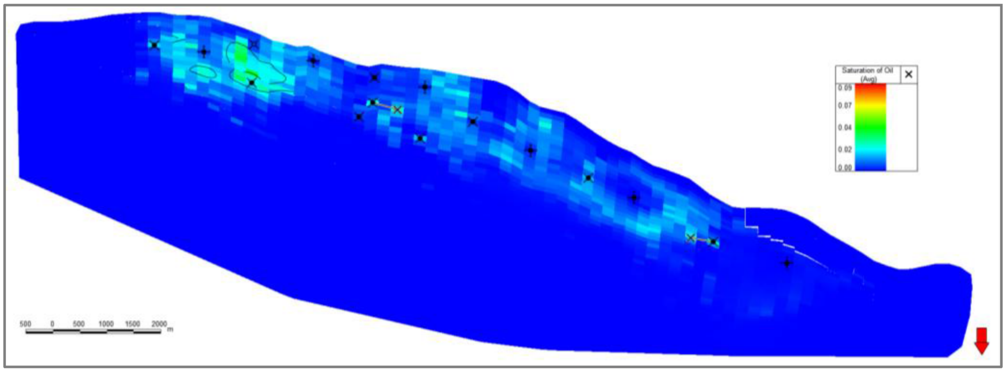

Figure 6—Condensate Oil Saturation Distribution in Y Gas Reservoir after N2 injection

This integrated process yields two significant benefits for hydrocarbon composition. Firstly, it enhances condensate oil recovery, with simulations predicting an increase by approximately 8 percentage points (Fig. 7). Secondly, it effectively displaces natural gas in the reservoir with nitrogen, leading to a substantial increase in natural gas recovery, estimated at over 10%. This innovative approach achieves dual revenue streams: from energy storage and from enhanced oil and gas production. The proposed multi-step process, particularly the strategic use of nitrogen injection prior to air storage, represents a sophisticated approach that simultaneously addresses a critical safety concern (oxygen-hydrocarbon interaction) and creates substantial economic value through enhanced hydrocarbon recovery. This synergistic strategy transforms a potential operational liability (residual hydrocarbons) into a significant asset. By proactively reducing hydrocarbon content to safe levels while recovering valuable resources, the project significantly improves its overall economic viability and de-risks the subsequent air storage operations. This is a key differentiator that adds unique value to the concept of repurposing condensate gas reservoirs.

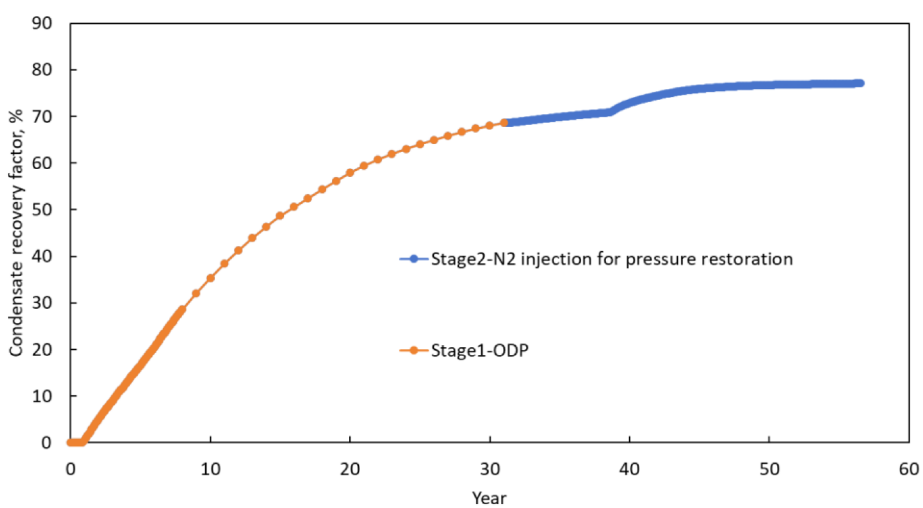

Figure 7—Comparison of condensate recovery between ODP and N2 injection for pressure restoration

Expected Economic Benefits

The assumptions for economic parameters are summarized in Table 1. Based on the investigation of the compressed air energy storage project, the initial investment for the project is estimated at 7.3 billion USD. The surface facilities investment, including gas injection facilities, power generation facilities, and grid connection, accounts for approximately 70% (5.1 billion USD). The underground investment, including new drilling, workover of existing wells, and salt cavern preparation, accounts for about 30% (2.2 billion USD). Assuming one charge-discharge cycle per day and 330 operating days per year, the annual power generation is approximately 37.62 TWh. The core revenue of the energy storage project comes from the electricity price difference. In China, the peak-valley price gap ranges from 0.2 to 0.4 CNY/kWh, equivalent to 0.028–0.055 USD/kWh. For economic evaluation, a value of 0.035 USD/kWh is adopted. The system efficiency is set at 0.72, slightly higher than the 70% efficiency level of current demonstration salt cavern CAES projects, but achievable through advanced thermal management technology. The annual operation and maintenance (O&M) cost for energy projects typically ranges from 1% to 2% of the investment; here, 1.8% is used. A discount rate of 6% and salvage value of 5% are used, consistent with standard parameters for energy projects. The expected lifespan of the project is 30 years. It is important to note that, for this primary economic assessment, the increased revenue from natural gas and condensate oil production (as discussed in above section) is temporarily not considered, providing a conservative financial outlook.

Table 1—Economic Assessment of Converting Depleted Gas Reservoirs to CAES

Parameter	Value	Unit
Initial Investment	7.3	Billion USD
Annual Discharge Volume	330×114GWh=37.62TWh	TWh
Electricity Price Difference	0.035	USD/kWh
System Efficiency	0.72	
O&M Cost	Initial Investment ×1.8%	Billion USD
Discount Rate	6	%
Salvage Value	5	%
Expected Lifespan	30	Years

Utilizing a full lifecycle economic evaluation model, the following key financial metrics are derived: a Net Present Value (NPV) of approximately 10.0 billion USD, an Internal Rate of Return (IRR) of approximately 14.3% and a Payback Period of 6.8 years.

A sensitivity analysis reveals that the project's economic viability is particularly sensitive to fluctuations in electricity price policies and the overall energy storage scale. A ±20% fluctuation in electricity price policy can lead to a significant 25-32% change in Internal Rate of Return (IRR). Furthermore, for every 10% decrease in working gas volume or energy storage scale, the Internal Rate of Return (IRR) is projected to decrease by 1.5 percentage points. The high sensitivity of the project's economics to external factors like electricity price policies and the achievable storage scale underscores that while the project is technically feasible and economically promising, its ultimate success is heavily reliant on a stable and supportive regulatory and market environment. This highlights the critical role of government policies (e.g., favorable peak-valley pricing, carbon market mechanisms) in de-risking large-scale energy infrastructure investments and ensuring their long-term profitability. It also stresses the importance of precise geological characterization and accurate volumetric assessment to minimize uncertainties in project scale.

Conclusions

Research on using gas or condensate gas reservoirs for CAES is scarce. This paper provides a foundational analysis of utilizing depleted oil and gas reservoir space for sustainable resource management, examining geological characteristics, energy storage scale, and economic benefits. Key findings include:

1. Converting condensate gas reservoirs to CAES offers geological feasibility, technical efficiency, and economic profitability. They provide excellent natural sealing, large storage capacities, and high energy densities. Strategic gas injection and nitrogen displacement enhance natural gas and condensate oil recovery, creating a unique dual revenue model.
2. In regions rich in gas and condensate gas reservoirs, the theoretical total energy storage capacity is immense. Integrating these facilities with UHV transmission networks could establish regional clean energy dispatch centers, addressing renewable energy instability. This repurposing strategy enables sustainable long-term utilization of depleted reservoirs with promising prospects.
3. Successful conversion requires overcoming a "geology-safety-system" technical barrier. Long-term reservoir stability needs further research via high-frequency injection-production experiments and stress field studies. Safety requires defining dual thresholds for oxygen and methane content. System optimization necessitates advanced simulation experiments for various energy flows, both above and below ground.

Acknowledgment

Acknowledgement Sincere thanks to New Mechanisms and Methods for Enhancing Recovery in Condensate Gas Reservoirs (2023ZZ0406) for their support and for granting permission to publish the paper.

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